

Install nfs

```
post1@post1:~$ sudo exportfs -v
/mnt/nfs_share 192.168.232.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw,insecure,root_squash,no_all_squash)
post1@post1:~$
```

```
post1@post1:~$ sudo systemctl status nfs-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: active (exited) since Fri 2026-07-03 12:50:20 UTC; 4min 18s ago
   Main PID: 3098 (code=exited, status=0/SUCCESS)
   CPU: 12ms
```

Xem kết quả trên máy 2

```
post2@post2:~$ sudo df -h
Filesystem                                Size  Used Avail Use% Mounted on
tmpfs                                     387M  1.6M  386M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv        9.8G  3.1G  6.3G  33% /
tmpfs                                     1.9G  1.1M  1.9G   1% /dev/shm
tmpfs                                     5.0M   0    5.0M   0% /run/lock
/dev/sda2                                1.8G  104M  1.6G   7% /boot
tmpfs                                     387M  12K  387M   1% /run/user/1000
192.168.232.13:/mnt/nfs_share             9.8G  2.9G  6.4G  31% /mnt/nfs_share
post2@post2:~$
```

Cài MinIO Server

```
post1@post1:~$ sudo groupadd -r minio-user
[sudo] password for post1:
post1@post1:~$ sudo useradd -M -r -g minio-user minio-user
post1@post1:~$ sudo mkdir -p /mnt/minio-data
post1@post1:~$ sudo chown minio-user:minio-user /mnt/minio-data
post1@post1:~$ curl -L -o minio https://dl.min.io/server/minio/release/linux-amd64/minio
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
ent                                     Dload  Upload  Total  Spent    Left  Spee
d
 0     0     0     0     0     0      0      0 --:--:-- --:--:-- --:--:--
 0     0     0     0     0     0      0      0 --:--:-- --:--:-- --:--:--
100 147 100 147     0     0  127      0  0:00:01  0:00:01 --:--:-- 1
27
 0     0     0     0     0     0      0      0 --:--:--  0:00:01 --:--:--
 0     0     0     0     0     0      0      0 --:--:--  0:00:01 --:--:--
 0
10 105M 10 10.7M     0     0 4135k      0  0:00:26  0:00:02  0:00:24 413
51 105M 51 54.7M     0     0 14.9M      0  0:00:07  0:00:03  0:00:04 44.
99 105M 99 104M     0     0 22.3M      0  0:00:04  0:00:04 --:--:-- 45.
100 105M 100 105M     0     0 22.4M      0  0:00:04  0:00:04 --:--:-- 45.
9M
post1@post1:~$ chmod +x minio
post1@post1:~$ sudo mv minio /usr/local/bin/
```

Cấu hình và tạo systemd service

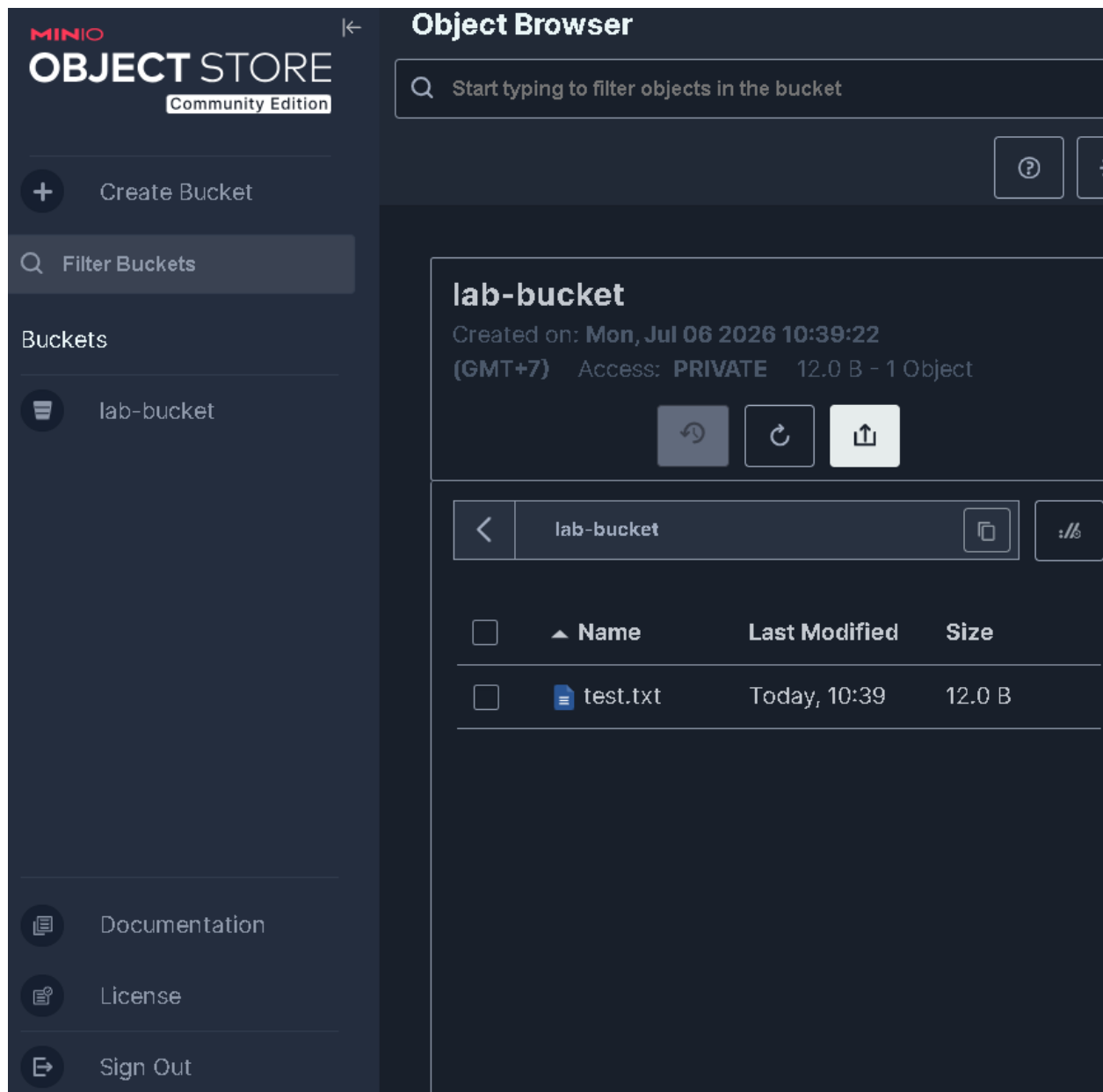
```
post1@post1:~$ sudo tee /etc/systemd/system/minio.service > /dev/null <<'EOF'
[Unit]
Description=MinIO Object Storage
Wants=network-online.target
After=network-online.target
AssertFileIsExecutable=/usr/local/bin/minio

[Service]
Type=notify
User=minio-user
Group=minio-user
EnvironmentFile=/etc/default/minio
ExecStart=/usr/local/bin/minio server $MINIO_VOLUMES $MINIO_OPTS
Restart=always
RestartSec=5
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
EOF
```

```
post1@post1:~$ sudo systemctl enable --now minio
Created symlink /etc/systemd/system/multi-user.target.wants/minio.service →
/etc/systemd/system/minio.service.
post1@post1:~$ sudo systemctl status minio
● minio.service - MinIO Object Storage
   Loaded: loaded (/etc/systemd/system/minio.service; enabled; preset: en>
   Active: active (running) since Mon 2026-07-06 03:35:24 UTC; 3s ago
     Main PID: 1733 (minio)
        Tasks: 8 (limit: 6955)
      Memory: 66.5M (peak: 66.8M)
         CPU: 626ms
       CGroup: /system.slice/minio.service
              └─1733 /usr/local/bin/minio server /mnt/minio-data --console-a>
```

đăng nhập bằng minioadmin



Cài MinIO Client (mc)

```

post2@post2:~$ sudo curl -L https://dl.min.io/client/mc/release/linux-amd64/
mc -o /usr/local/bin/mc
[sudo] password for post2:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
ent                                     Dload  Upload  Total  Spent    Left     Spee
d
  0      0     0     0     0     0      0      0  --:--:-- --:--:-- --:--:--
  0      0     0     0     0     0      0      0  --:--:-- --:--:-- --:--:--
100  141  100  141     0     0   137      0  0:00:01 0:00:01 --:--:-- 1
37
  0      0     0     0     0     0      0      0  --:--:-- 0:00:01 --:--:--
  0      0     0     0     0     0      0      0  --:--:-- 0:00:01 --:--:--
  0
  1 29.1M    1  479k     0     0  227k      0  0:02:11 0:00:02 0:02:09 22
 94 29.1M   94 27.4M     0     0 9018k      0  0:00:03 0:00:03 --:--:-- 26.
100 29.1M  100 29.1M     0     0 9393k      0  0:00:03 0:00:03 --:--:-- 27.
0M
post2@post2:~$ mc alias set local http://192.168.232.13:9000 minioadmin 1234
56789
mc: Configuration written to `/home/post2/.mc/config.json`. Please update yo
ur access credentials.
mc: Successfully created `/home/post2/.mc/share`.
mc: Initialized share uploads `/home/post2/.mc/share/uploads.json` file.
mc: Initialized share downloads `/home/post2/.mc/share/downloads.json` file.
Added `local` successfully.
post2@post2:~$ mc admin info local
• 192.168.232.13:9000
  Uptime: 3 minutes
  Version: 2025-09-07T16:13:09Z
  Network: 1/1 OK
  Drives: 1/1 OK
  Pool: 1

```

Pool	Drives Usage	Erasure stripe size	Erasure sets
1st	31.9% (total: 9.2 GiB)	1	1

```

1 drive online, 0 drives offline, EC:0

```

Tạo User

```
post2@post2:~$ mc admin user add local labuser 123456789
Added user `labuser` successfully.
post2@post2:~$ mc admin user info local labuser
AccessKey: labuser
Status: enabled
PolicyName:
MemberOf: []
post2@post2:~$ mc admin user list local
enabled      labuser
post2@post2:~$ mc mb local/lab-bucket
```

Upload Object

```
post2@post2:~$ echo "Hello MinIO" > test.txt
mc cp test.txt local/lab-bucket/
.../test.txt: 12 B / 12 B | ████████████████████████████████████████████████████████████ 747 B/s 0s
post2@post2:~$ mc cp --recursive ~/demo-folder/ local/lab-bucket/demo-folder
/
mc: <ERROR> Unable to prepare URL for copying. Object does not exist
post2@post2:~$ mc ls local/lab-bucket
[2026-07-06 03:39:30 UTC]      12B STANDARD test.txt
post2@post2:~$
```

lab-bucket

Created on: **Mon, Jul 06 2026 10:39:22**
(GMT+7) Access: **PRIVATE** 12.0 B - 1 Object

☐	Name	Last Modified	Size
☐	 test.txt	Today, 10:39	12.0 B

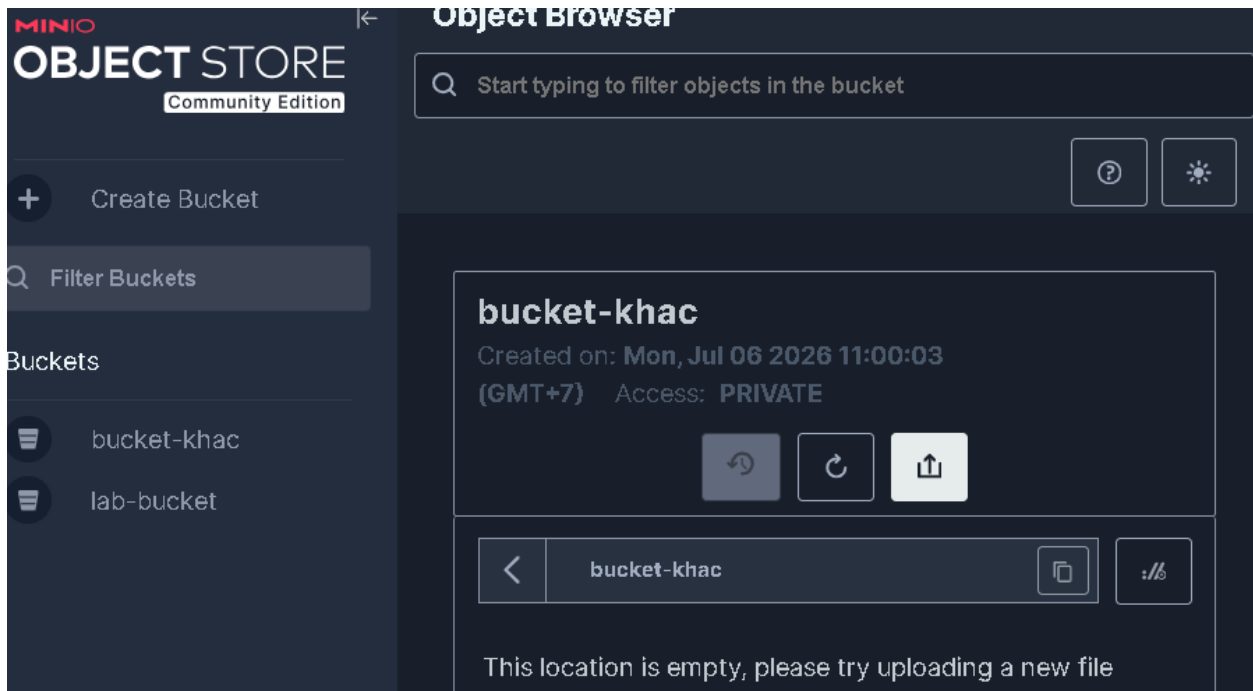
Set nhanh quyền truy cập bucket

```
post2@post2:~$ mc anonymous set download local/lab-bucket
Access permission for 'local/lab-bucket' is set to 'download'
post2@post2:~$ mc anonymous set none local/lab-bucket
Access permission for 'local/lab-bucket' is set to 'private'
post2@post2:~$ mc anonymous get local/lab-bucket
Access permission for 'local/lab-bucket' is 'private'
```

```
post2@post2:~$ mc admin policy attach local readonly --user labuser
Attached Policies: [readonly]
To User: labuser
post2@post2:~$ mc admin policy entities local --user labuser
Query time: 2026-07-06T03:56:10Z
User -> Policy Mappings:
  User: labuser
  Policies:
    readonly
```

```
post2@post2:~$ mc admin policy entities local --user labuser
Query time: 2026-07-06T03:56:10Z
User -> Policy Mappings:
  User: labuser
  Policies:
    readonly
post2@post2:~$ cat > lab-policy.json <<'EOF'
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:GetObject", "s3:PutObject", "s3:ListBucket"],
      "Resource": [
        "arn:aws:s3:::lab-bucket",
        "arn:aws:s3:::lab-bucket/*"
      ]
    }
  ]
}
EOF
post2@post2:~$ mc admin policy create local lab-bucket-policy lab-policy.json
Created policy 'lab-bucket-policy' successfully.
post2@post2:~$ mc admin policy attach local lab-bucket-policy --user labuser
Attached Policies: [lab-bucket-policy]
To User: labuser
```

Test đăng nhập minio admin tạo 1 bucket khác



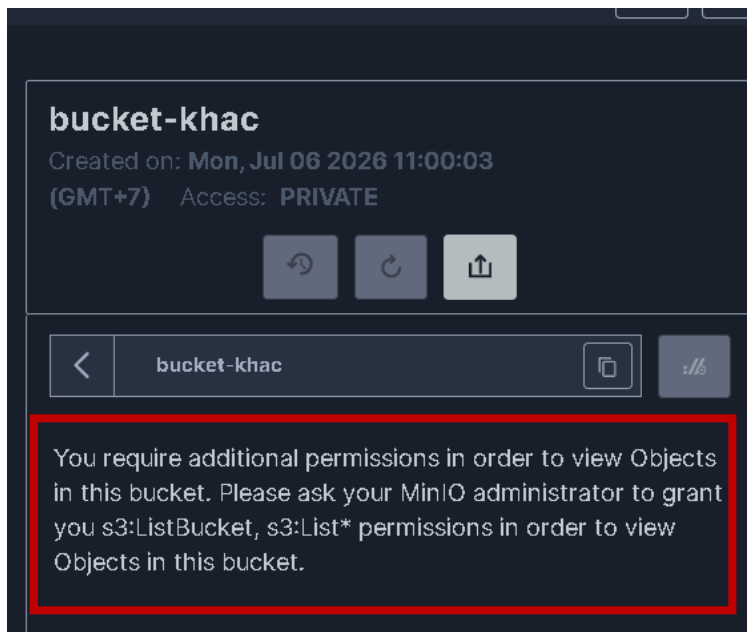
đăng nhập bằng labuser test quyền

test trên API s3

```
post2@post2:~$ mc alias set labuser-session http://192.168.232.13:9000 labuser 123456789
Added 'labuser-session' successfully.
post2@post2:~$ mc ls labuser-session/lab-bucket
[2026-07-06 03:39:30 UTC]    12B STANDARD test.txt
post2@post2:~$ mc mb labuser-session/bucket-khac
mc: <ERROR> Unable to make bucket 'labuser-session/bucket-khac'. Access Denied.
post2@post2:~$
```

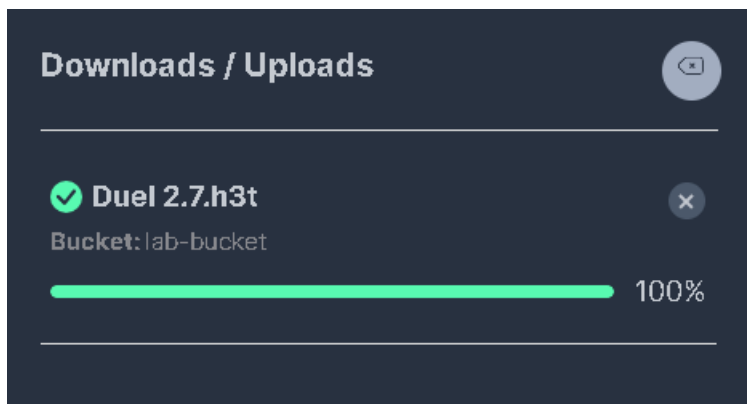
test trên web

Không có quyền xem hay truy cập vào bucket khác

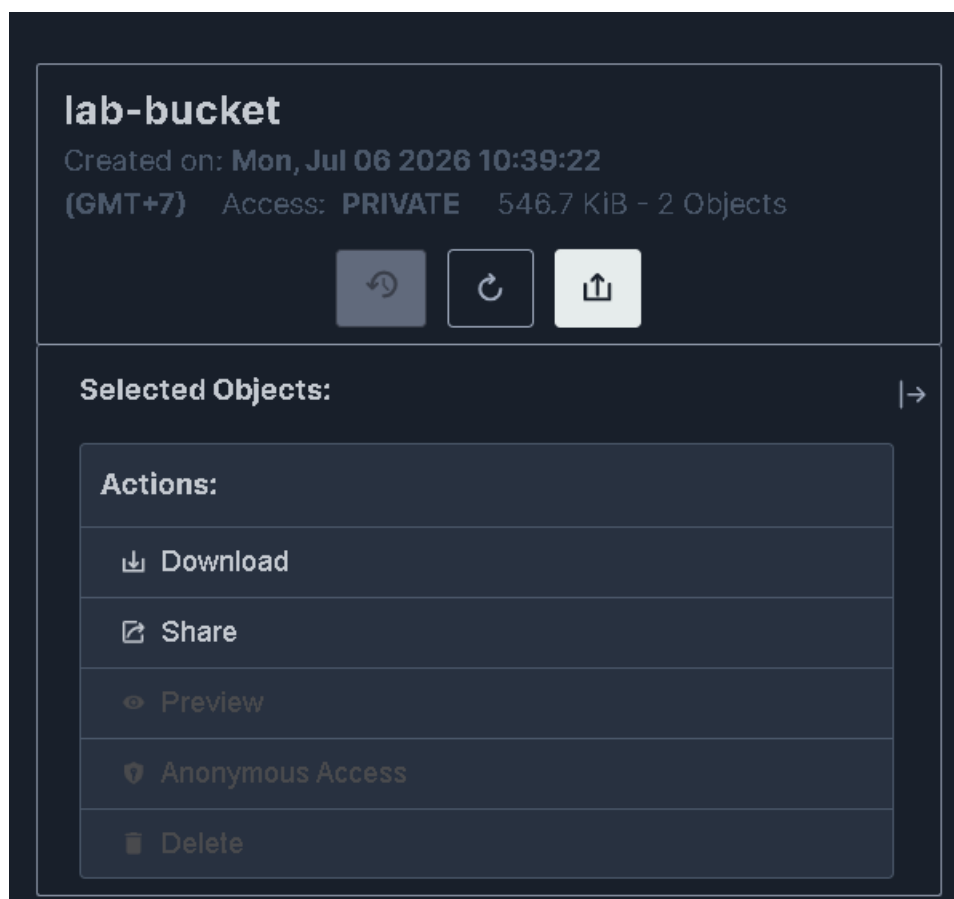


Cho phép upload tài liệu tại lab-bucket

Upload project done!



Cho phép download tài liệu



Chặn labuser không cho phép up/download

Gỡ lab-bucket-policy

```
post2@post2:~$ mc admin policy detach local lab-bucket-policy --user labuser
Detached Policies: [lab-bucket-policy]
From User: labuser
```

```
Detached Policies: [lab-bucket-policy]
From User: labuser
post2@post2:~$ mc admin user info local labuser
AccessKey: labuser
Status: enabled
PolicyName: readonly
MemberOf: []
post2@post2:~$
```

Lúc này labuser chỉ xem đc không thao tác đc trên lab-bucket

